Ordinary Differential Equations: Abridged Notes

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These abridged notes were reconstructed afterward because I did not take direct notes during class. They are not a record of the actual lecture.

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1 Topic Summary

For a homogeneous second-order linear differential equation with constant coefficients,

$$ay'' + by' + cy = 0, \qquad a \neq 0,$$

trying $y = e^{rt}$ turns the differential equation into the characteristic equation $ar^2 + br + c = 0$. Its roots determine a fundamental pair of solutions. Distinct real roots give two distinct exponentials; repeated and complex roots require different forms. Initial conditions determine the constants in the general solution.

2 The Characteristic Equation

If r_1 and r_2 are distinct real roots of $ar^2 + br + c = 0$, then

$$\boxed{y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}.}$$

The two exponentials are linearly independent because their Wronskian is $(r_2 - r_1)e^{(r_1+r_2)t} \neq 0$. The form of the solution changes for a double root or a conjugate pair; those cases are treated in later notes.

2.1 Example: Two Distinct Real Roots

Adapted from Boyce et al., 11th ed., §3.1, Problem 1. Solve $y'' + 2y' - 3y = 0$.

Solution.

The characteristic equation factors as

$$r^2 + 2r - 3 = (r - 1)(r + 3) = 0.$$

The roots are 1 and -3, so

$$\boxed{y(t) = C_1 e^{t} + C_2 e^{-3t}.}$$

Each exponential satisfies the equation, and their Wronskian $-4e^{-2t}$ is nonzero.

2.2 Example: An Initial-Value Problem and Minimum

Adapted from Boyce et al., 11th ed., §3.1, Problem 14. Solve $y'' - y = 0$, $y(0) = 5/4$, $y'(0) = -3/4$; find its minimum for $0 \leq t \leq 2$.

Solution.

The characteristic roots are 1 and -1, so $y = C_1 e^t + C_2 e^{-t}$. At $t = 0$, the data give

$$C_1 + C_2 = \frac{5}{4}, \qquad C_1 - C_2 = -\frac{3}{4}.$$

Thus $C_1 = 1/4$, $C_2 = 1$, and

$$y(t) = \frac{1}{4}e^t + e^{-t}.$$

Differentiating gives $y' = e^t/4 - e^{-t}$. Its only critical point is $t = \ln 2$, which lies in $[0, 2]$. Since $y'' = e^t/4 + e^{-t} > 0$, this point is a minimum, with

$$\boxed{y(\ln 2) = 1.}$$